

Deepak Mallampati

+1-330-554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati |
Kent, OH (US-based, F-1 OPT, open to sponsorship)

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) who ships internal AI solutions end-to-end - diagnose the workflow first, then build with agentic architecture, multi-agent orchestration, prompt engineering with system-prompt + context-management discipline, and guardrails. Solo-shipped Manga Lens to the Chrome Web Store integrating four AI vision providers (Anthropic Claude, OpenAI GPT-4o mini + GPT-4.1 Nano, Ollama OSS). Production: Healthcare RAG (~35% retrieval precision gain), agentic LLM workflows (>30% hallucination reduction), schema-validated multi-agent claims pipelines with audit-ready reasoning traces, and FastAPI / Docker production discipline (~30% post-deployment defect reduction).

SKILLS

Agentic Architecture & Multi-Agent Orchestration • Prompt Engineering & Context Management • Multi-Provider LLM Integration (Anthropic / OpenAI / OSS) • Guardrails & AI Safety • Python / TypeScript / REST APIs • Production Workflow Automation • Stakeholder Diagnosis • CI/CD Discipline

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, JavaScript (MV3), C++ (Arduino), PHP

LLM & GenAI: LLM application development, agentic architecture (tool use, multi-agent orchestration, guardrails), RAG, structured outputs, prompt engineering, context management, evaluation pipelines, grounding, semantic retrieval, embeddings, vector search, responsible AI

LLM Platforms: Anthropic Claude (Sonnet via Manga Lens), OpenAI (GPT-4o mini + GPT-4.1 Nano), Open-source (Ollama / minicpm-v local), Hugging Face Transformers / Diffusers

ML Libraries: scikit-learn, XGBoost, PyTorch, LangChain, LlamaIndex

Integration & APIs: RESTful APIs, multi-provider API orchestration, JSON contracts, structured logging, webhook patterns

Infra & DevOps: Docker, Jenkins CI/CD, Grafana, MLOps/LLMOps, cloud-ready deployment, observability

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - USA (Healthcare Technology, AI Consulting)

- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care workflow navigation, documentation clarification) with structured reasoning, **tool discipline**, and **grounding rules**; agent response stability +~25%, hallucination reduction >30%.
- Built **RAG systems** for clinical knowledge retrieval; recursive semantic chunking + transformer embeddings; contextual retrieval precision +~35%, grounded-response alignment >90%.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, structured logging + load simulation; post-deployment defects ~30%.
- Wrote stakeholder-facing system-limitation docs alongside every release - the artifact non-technical clinicians and ops rely on (diagnose-first principle in action).
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost); recall +15-20% on high-risk categories at precision >90%.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Hyderabad, India

- Built **CI/CD pipelines with Jenkins** for schema updates with rollback checkpoints; deployment errors ->30%, release cycle -35-40%.
- Designed **SQL Server DMVs + Grafana** performance dashboards; incident recurrence ~25%, root-cause resolution +~30%.

Machine Learning Engineer - Hyderabad, India

- Built a **transformer-based video summarization**; review time **-60-70%**, ~85% highlight alignment.
- Implemented **document Q&A with RAG**; hallucinated outputs **~30%**, answer accuracy **>85%**.

PROJECTS

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation (shipped, Chrome Web Store)

Full-stack browser extension built independently - Manifest V3, content scripts, service workers. Integrates **four AI vision providers** (Anthropic Claude Sonnet, OpenAI GPT-4o mini, OpenAI GPT-4.1 Nano, Ollama/minicpm-v local OSS) with per-provider payload handling (WebP for cloud, JPEG for Ollama to avoid CUDA crash). 7-day cache; 29 site configs; privacy policy; narrowed host permissions. **Direct Anthropic + OpenAI + OSS multi-provider integration evidence.**

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake → Validation → Consistency → Duplicate → Risk-Scoring) with **schema-validated JSON contracts between agents** - direct multi-agent orchestration + guardrail pattern. RAG-grounded CPT/ICD validation; ANN duplicate detection; explainable risk scoring with audit-ready reasoning traces.

Dream Decoder - AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app coordinating multimodal APIs. **Intermediate structured prompt transformation layers** between APIs improved contextual alignment **~30%** and first-pass image success rate **~25-30%**.

Driver Drowsiness Detection - Real-Time YOLOv8 Fatigue Monitoring

Replaced a two-stage CNN pipeline with a unified **YOLOv8** model; inference latency **~30%**; false positives **~25%**.

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025